Penmetsa Srikar Varma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.13
FIITJEE Junior College, Visakhapatnam	Class XII (Andhra Pradesh State Board)	2023	96.50%
Chegu Vidyalayam	Class X (Andhra Pradesh State Board)	2021	100.00%

WORK EXPERIENCE

Digital IC Engineer | *Deepgrid Semiconductor, Hyderabad*

- Reduced 75ms inference latency upon optimizing pipeline for Ele-mul, Maxpool, Transpose operators in YOLO model
- Re-developed Mac and Ele-mul operators pipeline supporting fp8, bf16 and fp4 multi-precisions at hardware level
- Architect-ed a wrapper design for ASIC-Attention engine enabling Softmax operator block on FPGA level

ASIC Layout Design | *DRDO Project* | ***Prof. Gajendranath Chaudhury***

- Designed layout for 5V-5T OTA and Temperature Sensor of Gyroscope Readout IC using TSMC 180nm technology node
- Gained hands-on experience on Circuit design, Placement, Routing with DRC, LVS checks in Cadence Virtuoso

PROJECTS

DFT Based Digital ASIC Flow (In-progress) | Tools: *Cadence Genus, Innovus, Xilinx Vivado, Vitis*

- Implemented and verified a MAC DUT with BRAMs on ZCU104 FPGA using Vivado Vitis HW-SW application platform
- Achieved 99.1%+ ATPG based Fault-coverages in each of Stuck-at, Transition, Bridging faults delivering .stil patterns file
- Executing complete RTL-GDSII flow for verified MAC with DFT in TSMC 40nm Technology node delivering GDSII file

RTL Design and Verification (In-progress) | Tools: *Icarus Verilog, Vivado*

- Develop and verifying a scheduler RTL wrapper for multi-stage Softmax pipeline with inter-stage buffering, FSM-based inter-tile handshakes and validated the design functionality using testbench

RTL Benchmark Synthesis | Tools: *Synopsys Genus, Bash script, Jupyter notebook*

- Executed RTL synthesis for 10 benchmark designs with Tcl script and Sdc constraints
- Performed synthesis across frequency libraries 100MHz, 400MHz and 6 different Vth technology libraries
- Generated Power, Area, Timing, Netlist reports and performed block-wise Power & Area analysis

FPGA Implementation Testing | Tools: *iverilog, gtkwave*

- Designed a Digital clock (in Verilog) and implemented on Vaman FPGA using LCD interface monitoring
- Generated FPGA flashable *bin file (design and testbench) and *pcf files using design files in Remote server

High-gain Amplifier Design | Tools: *Open-source, NGSPICE*

- Designed single-stage CS and cascode-stage amplifiers using IIC-OISC-TOOLS LVMOS IHP SG132 PDK
- Validated the design using NGSPICE simulation results achieving specs and specs documenting the final netlist

High-gain OTA Design | Tools: *Open-source, NGSPICE*

- Designed Two-stage Miller compensated OTA on 130nm BiCMOS PDK using open-source IIC-OISC-TOOLS
- Validated the design using NGSPICE simulation results achieving specs documenting the final netlist

Semiconductor Device Modeling | Tools: *C, Python, Jupyter notebook*

- Modeled steady-state PN Junction-diode Carrier concentration profiles using python
- Demonstrated a Non-uniform Mesh analysis for Poisson solver, Carrier continuity equations in C language

Custom PCB Design | Tools: *KiCad, Solid Edge, Altium Designer*

- Designed a 4-layer multilayer PCB (in KiCad), CAD model (in Solid edge) for Electronics systems design lab
- Designed 2 custom PCBs using Renesas-built DA14531 IC (in Altium designer) problem statements

SKILLS

HDL - Verilog, System Verilog

Programming Languages - C, Python

VLSI Tools & Scripts - Unix shell, Tcl script, Cadence Virtuoso, Xilinx Vivado, LTSpice, KiCad, Solid Edge

Microcontroller Boards - Xilinx Artix, Xilinx ZCU104, Vaman FPGA

RELEVANT COURSES

Digital VLSI - Digital Electronics and Lab, Digital IC Design, STA, RTL Design and Verification

Analog VLSI - Analog Electronics and Lab, Analog IC Design, Altium PCB Design

Other Courses - Computer Architecture, Control Systems, Semiconductor Device Fundamentals, Embedded Systems

POSITIONS OF RESPONSIBILITY

Teaching Assistant | *Digital Systems Lab* | ***Prof**. Gajendranath Chaudhury*

- Deployed Assignments, Quizzes, Final project - Problem statements and conducted Viva for evaluation
- Delivered Theory and Lab sessions on Verilog HDL, guiding total of 90+ students

Internship Coordinator | *Office of Career Services*

- Coordinated B.Tech internships through pitching and engaging with industrial HR professionals
- Provided internship opportunities for B.Tech students by being part of OCS Team, IIT Hyderabad

EXTRACURRICULAR ACTIVITIES

- **NSO, IIT Hyderabad** - Achieved contact hours by engaging in cricket as a bowler

- Hobbies include Cricket, Long jump, Athletics, Badminton and Carrom